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Smart Waste Collection Based on Vehicle Routing Optimization: Case of Casablanca City

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Abstract

Morocco has implemented circular economy principles in their waste management program to close or rehabilitate all existing landfills. However, the waste collection and transportation process remain highly inefficient and costly, contributing to a substantial environmental footprint for waste collection operations. The inefficiencies in the current waste management system highlight the need for innovative solutions to optimize collection processes and minimize environmental impact. This study responds to this imperative by proposing the use of smart bin technology combined with a vehicle routing problem tool. The integration of these technologies aims to develop an intelligent solution for appropriate waste collection, limiting container pickup only when necessary, and optimizing waste collection routes. Our approach considers several factors, including the fill level of each container, time windows, vehicle capacity, and tour completion constraints. By addressing these key factors, the system seeks to revolutionize waste management practices, improving the efficiency and effectiveness of waste collection.

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Keywords: **Smart Waste Management, Vehicle Routing Problem with Time Windows (VRPTW), Internet of Things (IOT), Sustainable Waste Collection, Circular economy.**

1. Introduction

As urban centers experience escalating population growth, effective waste management emerges as a pivotal facet in

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upholding a sustainable and salubrious environment. Conventional waste collection procedures confront a gamut of challenges, encompassing the propensity for bin overflows and the exigency for judicious route optimization. Nevertheless, the advent of IoT technology [1] is revolutionizing waste management by furnishing real-time insights into the status of waste receptacles. This paradigm shift not only serves to curtail the incidence of uncollected or overflowing garbage receptacles but also streamlines resource allocation. Furthermore, it contributes to the abatement of fuel consumption and the minimization of temporal inefficiencies, thus augmenting the overall operational efficiency of waste management [2].

Several companies have adopted IoT technology to revolutionize waste management, thereby creating an intelligent system that offers a plethora of high-quality data, including information on location, fill levels, temperature, humidity, and the ability to alert stakeholders in the event of an incident [3] [4] [5].

To enhance their efficacy, these data need to be integrated and communicated for the purpose of strategically planning efficient collection routes, resulting in fuel and labor savings.

This project aims to introduce a novel system that integrates a smart dustbin with a vehicle routing tool to improve the waste collection process. The smart dustbin will be equipped with IOT sensors that will constantly monitor the level of waste inside the bin. The vehicle routing tool will use this information to generate an optimal itinerary that can efficiently collect full bins before they become overloaded and prevent the unnecessary collection of empty bins.

The remaining sections of this paper are structured as follows: Section 2 delves into previous research efforts exploring the utilization of technologies for enhancing collection and transport phases. Section 3 is dedicated to outlining the methodology employed in this study, while Section 4 comprehensively showcases the results obtained, accompanied by a focused discussion on these findings.

2. Related Work

In recent years, researchers have proposed innovative solutions within the realm of smart systems to address urban challenges worldwide. Notably, these solutions have been directed towards the enhancement of smart waste management, particularly targeting issues related to waste collection and vehicle routing.

For instance, a noteworthy development by [6] introduced a smart garbage bin furnished with an Ultrasonic Sensor, Bluetooth capability, and a Direct Current (DC) motor. This advanced bin autonomously monitors waste levels, compacts waste, and can be controlled through an Android application.

In another research endeavor [7], the use of ultrasonic level sensors, an Arduino Nano board, and a GPS communication module allowed for real-time monitoring of garbage can capacity. Stakeholders are notified when necessary, and usage and filling events are recorded on a memory card through the utilization of a Passive Infrared (PIR) sensor.

An additional study [8] presented an efficient control mechanism involving infrared (IR) sensors, a transmitter, a microcontroller, and a radio frequency (RF) module to provide real-time monitoring of solid waste levels in garbage cans. These sensors accurately gauge waste levels and promptly communicate alerts to the control system.

Moreover, a different "smart bin" development [9] integrated a suite of sensors, including an accelerometer, temperature and humidity sensors, and an ultrasonic sensor. These sensors are managed by a Zigbee Pro microcontroller card, which is integrated with a Wi-Fi module to transmit data to a central gateway.

The implementation of smart bins offers an array of benefits, including enhanced waste management, reduced littering, and improved environmental sustainability. Additionally, research findings [10] suggest that the deployment of smart trash bin projects can have a positive influence on community attitudes towards proper waste disposal. Pre-project implementation, 69.2% of respondents held a favorable stance on proper waste disposal, a figure that increased to 76.6% after the introduction of the smart trash bin project.

Waste transportation carries notable environmental implications, contributing around 30% of global emissions [11]. The associated financial costs can be substantial and may rise due to logistical issues [12].

The Vehicle Routing Problem (VRP) has evolved into a complex optimization challenge, with various algorithms developed to address it and its variants. VRP is essential for cost reduction in diverse domains, defining least-cost routes for vehicles serving dispersed customers with known demand [13]. It entails a depot and vehicles with capacity constraints, each serving customers once per trip without exceeding demand, returning to the depot.

Numerous VRP versions and solutions have emerged, often necessitating heuristics and metaheuristics due to

complexity. Prominent algorithms include VRP, Captive Arc Routing Problem (CARP), and their derivatives [14] [15]. In a study in Troyes, France [16], the Capacitated Arc Routing Problem (CARP) efficiently minimized total distance travelled and longest distance.

In a study [17], a waste collection model leveraging real-time traceability data optimizes routes, considering factors like bin levels, waste types, and vehicle locations. The proposed multi-objective optimization model, tested in an Italian system, effectively reduces costs, and minimizes environmental impact. The research highlights the benefits of using traceability data and advanced models in waste management.

To tackle the issue of solid waste collection in Onitsha city, Nigeria, a solution was devised using an extended Captive Arc Routing Problem (CARP) on an undirected network. The employed approach heuristic [18], follows a two-step process: first establishing routes and then clustering.

This allowed to exclude a vehicle from use, as well as saving 16.31% of route travelled, 25.24% in cost of collection and 23.51% in time travel.

While several research efforts have been undertaken to integrate the concept of a smart dustbin through various methods or to utilize VRPS tools solutions for optimizing collection routes, these endeavors have often remained incomplete. Therefore, this project aims to introduce a novel system that integrates a smart dustbin with a vehicle routing tool to enhance the waste collection process. The smart dustbin will be equipped with IoT sensors that continuously monitor the waste levels within the bin. The vehicle routing tool will utilize this information to generate an optimal route that efficiently collects full bins before they reach capacity, thus avoiding the unnecessary collection of empty bins.

3. Methodology

3.1. Collection and transport process in Casablanca

In Morocco, waste collection employs two primary modes: voluntary contribution, offering accessible containers, and door-to-door services, where assigned containers are used. Operations run daily (365 days), with collections at 9 p.m. in high-density neighborhoods and 7 a.m. in others, subject to seasonal and Ramadan-related schedule adjustments [19].

The waste collection cycle commences at the Mediouna landfill (as illustrated in Fig.1). Vehicles follow designated routes, collecting waste until they reach their capacity limit, at which point they unload at the landfill. This process repeats during operational hours, and all vehicles return to the landfill.

Casablanca's waste collection system operates continuously, although capacity limitations can impact truck efficiency.

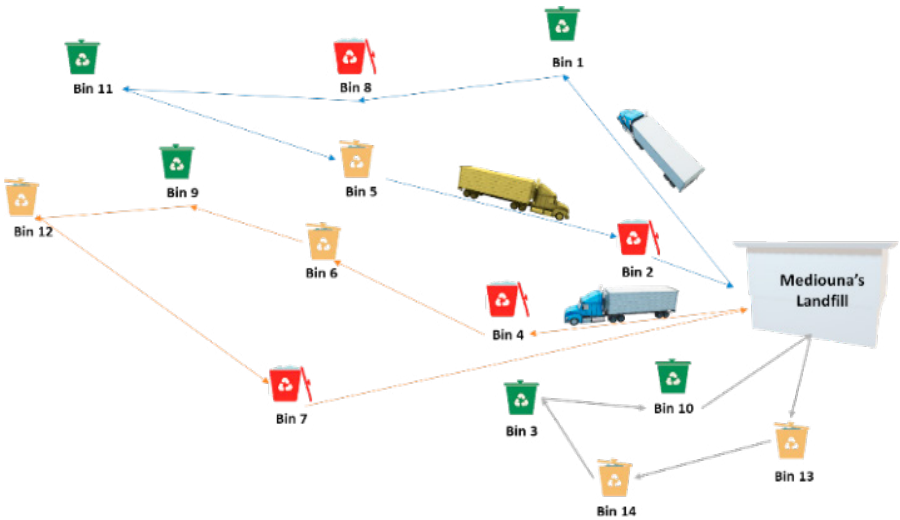

Fig. 1. The current waste collection process.

Optimizing waste collection with single-vehicle bin assignments boosts efficiency, cuts costs, and lessens environmental impact. Benefits include streamlined routes, reduced fuel consumption, lower maintenance costs, enhanced organization, error prevention, reduced overfill risks, and a diminished carbon footprint, leading to lower emissions and less air pollution.
The challenge lies in determining the optimal route for each vehicle while considering the waste quantity in each bin and adhering to specific prerequisites:

- Ensuring that the trucks do not exceed their load capacity.
- Designing routes that commence from the Mediouna's landfill, traverse the various waste locations, and return to the starting point.
- Optimizing the total distance traveled by each truck while maximizing the volume of waste collected.
- Ensuring that each waste bin is collected just once, and by a single designated vehicle.

3.2. The Proposed Smart Bin System

To address challenges and cost-efficiency in waste collection and transportation, a proposed smart waste bin aims to offer real-time monitoring of filling levels. This data will be integrated with the VRP tool to optimize collection routes. The proposed smart bin incorporates the following components:

- Arduino Mega 2560: Featuring 54 digital I/O pins, 16 analog inputs, 4 UARTs, and a 16 MHz crystal oscillator.
- Ultrasonic Sensors: Utilized for waste bin filling level detection, consisting of transmitters and receivers. Ultrasonic signals are transmitted, and reflections are received to gauge filling levels. These sensors operate on a 5V input.
- PIR Sensor: Detects human movement within a 10-meter radius through pyroelectric sensors sensitive to infrared radiation [20].
- ESP8266: A powerful Wi-Fi module offering internet connectivity and compatibility with microcontrollers, ideal for IoT applications.
- Real-Time Clock Module DS1302: A cost-effective, high-accuracy module providing time and date information, accounting for different month lengths and leap years.

To enable real-time monitoring of bin filling levels and usage status, PIR sensors continuously detect human presence, triggering the microcontroller to request filling level measurements from the ultrasonic sensor. Time information is sourced from the DS1302 Real-Time Clock module, while the Arduino Mega serves as the system's microcontroller, providing a 5V power supply. Data collected by the sensors is wirelessly transmitted via WIFI technology.
As public waste bins often have a capacity that exceeds typical waste accumulation, the ultrasonic sensor is positioned sufficiently above this threshold. This configuration allows the system to accurately assess both solid and liquid waste levels, as ultrasonic waves effectively reflect from liquid surfaces.

3.3. *VRP - Optimization Process*

The optimization process for various vehicle routing scenarios involves addressing specific constraints such as vehicle types (homogeneous, heterogeneous), demand characteristics (deterministic or stochastic), depot quantities, and road restrictions. Incorporating these constraints adds complexity and allows for the adaptation of VRP variants tailored to the specific context.
In waste management, arc routing problems are commonly linked to situations where strict time constraints are not a primary concern. This perspective is particularly relevant to household waste collection. The waste collection sector comprises three main segments: commercial waste collection, residential waste collection, and roll-on-roll-off. The following discussion centers on the challenges associated with commercial and residential waste collection, resembling a modified version of the Vehicle Routing Problem with Time Windows (VRPTW) due to the imposition of specific time frames for commercial waste collection stops.

First and to determine the optimized route, it was assumed that:

- The impact of traffic and any other delays encountered by a driver during the journey will be excluded.
- The conductor left the depot at 7:55 am and came back at 7:00 p.m.
- No time constraints at the Mediouna landfill.
- All commercial waste collection points open at 8:30 am and close at 5:30 p.m.
- The service duration at residential collection points is 5 min and 15 min at commercial ones.

3.4. *VRP – Open Door Logistic (ODL) tool*

ODL, built upon jsprit and OpenStreetMap, serves as an open-source fleet planner and territory management tool for transport schedulers and analysts [21]. It excels in solving VRP optimization problems, offers flexibility, user-friendliness, and free accessibility.
ODL supports route planning, automatic scheduling on real road networks with interactive maps, Gantt chart representation of vehicle usage times, and the generation of detailed reports in PDF format.
The Java-based "Jsprit" code provides adaptability for optimal problem-solving, allowing data editing within Excel tables. This interactive feature streamlines operations, supports analysis, report generation in PDF, or further study within Excel worksheets.
ODL enables the definition of model parameters and requirements, including truck capacities, locations, stop numbers, time windows, working hours, and associated costs. It combines these features and uses a large neighborhood algorithm for routing problem resolution.

The program configuration involved setting it to use the latest open card in the Moroccan area. Subsequently, input tables were created, resulting in the generation of "Stops," "VehicleTypes," and "Stops-order" tables in the worksheet. "Stops" and "VehicleTypes" contained incoming data, while output itineraries were stored in "Stops-order."

	id	address	latitude	longitude	quantity	type
1	Bin-is1	Bidonville Derb Ghall...	33,568	-7,63	345	D
2	Bin-is2	Val Fleurie	33,575	-7,637	435	D
3	Bin-is3	Rue A`t Bougamaz, C...	33,573	-7,659	363	D
4	Bin-is4	520 Rue Sidi Al Khadi...	33,571	-7,681	473	D
5	Bin-is5	RACINE EXTENSION,...	33,595	-7,641	500	D
6	Bin-is6	Ancienne Medina, Ca...	33,602	-7,621	220	D
7	Bin-is7	Avenue Ibn Zair Abde...	33,587	-7,563	436	D
8	Bin-is8	Hay Mohammadi	33,595	-7,556	124	D
9	Bin-is9	Boulevard Al Bina, Ca...	33,626	-7,511	353	D
10	Bin-is10	Boulevard de l'Océan ...	33,586	-7,672	361	D
11	Bin-is11	Boulevard d'Afghanist...	33,57	-7,681	123	D
12	Bin-is12	Boulevard Mohamed ...	33,583	-7,508	353	D
13	Bin-is13	Sidi Maarouf, Casabl...	33,518	-7,654	540	D
14	Bin-is14	Casanearshore Porte...	33,522	-7,641	300	D
15	Bin-is15	Promenade N, Bousk...	33,484	-7,584	580	D
16	Bin-is16	Préfecture de Moulay ...	33,522	-7,518	430	D
17	Bin-is17	Avenue 2 Mars, Casa...	33,531	-7,593	200	D

Fig. 2. Input data table – Bin (Quantity in kg).

The stop (Bin) table, as shown in Fig. 2, it will include the following data: Id, Name and address - *not used by optimizer,* Latitude and longitude - *bin location*, Quantity, *will be generated by the smart bin (in kg),* Required skills

or Type, *the appropriate truck should be capable of supporting the task.*

	vehicle-name	vehicle-id	start-latitude	start-longitude	end-latitude	end-longitude	start-time	end-time	capacity
1	Dump satellite dumper	DS-2424242	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	1 000
2	Dump satellite dumper	DS-2424243	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	1 000
3	Packer truck	PT8-2421431	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	600
4	Packer truck	PT8-2421432	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	600
5	Packer truck	PT8-2421433	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	600
6	Packer truck	PT8-2421434	33,49	-7,538	33,49	-7,538	07:55:00	19:00:00	600

Fig. 3. Input data table – VehicleTypes (Capacity in kg).

As indicated in Fig. 3, the prerequisites to define for the second input table VehicleTypes are: vehicle-name, vehicle-id, start-latitude, start-longitude – *Landfill location,* end-latitude, end-longitude - *Landfill location,* start-time - (07:55:00), end-time - (19:00:00), capacity, cost-per-km - cost-per-hour, skills - *meets the required skills in the stop table.*

4. Results

Once the ODL VRP tool is configured with input tables containing data on waste bins and available vehicles, launching the optimizer takes just 30 seconds. The result is a set of optimized routes, with assigned vehicles and collection points. The algorithm prioritizes factors like proximity, vehicle capacity, and bin fill levels to minimize travel time and vehicle count. This approach significantly enhances cost-efficiency and operational effectiveness in waste collection and transportation.

vehicle-name	vehicle-id	stops-count	travel-km	Picture
Dump satellite dumper	DS-2424242	5	52,195	

stop-number	stop-name	stop-address	arrival-time
1	Bin-is11	Boulevard d'Afghanistan,	08:15:10.326
2	Bin-is6	Ancienne Medina,	08:29:59.435
3	Bin-is3	Rue A¨t Bougamaz,	08:43:16.953
4	Bin-is2	Val Fleurie	09:02:38.614
5	Bin-is1	Bidonville Derb Ghallef,	09:10:11.854

Fig. 4 Itinerary followed by a dump satellite dumper.

The Fig. 4 provides a concrete example of the capabilities of the waste collection optimization system. The figure shows that a Dump satellite dumper waste truck, with a maximum load capacity of 1500 kg, was able to collect waste from five garbage cans of varying capacities in just 55 minutes, covering only 52 km and with only 380 MAD of costs. This optimized route plan resulted in significant time and fuel savings compared to traditional waste collection

methods. Furthermore, the optimization system enables efficient use of available resources. In this example, only six waste trucks are required to collect waste from all 31 garbage cans.

The ODL tool not only optimizes waste collection routes but also offers a valuable visualization tool via a Gantt chart (Fig. 5). This chart displays the start and end times of each waste truck's activities, facilitating real-time monitoring and identification of areas for improvement.

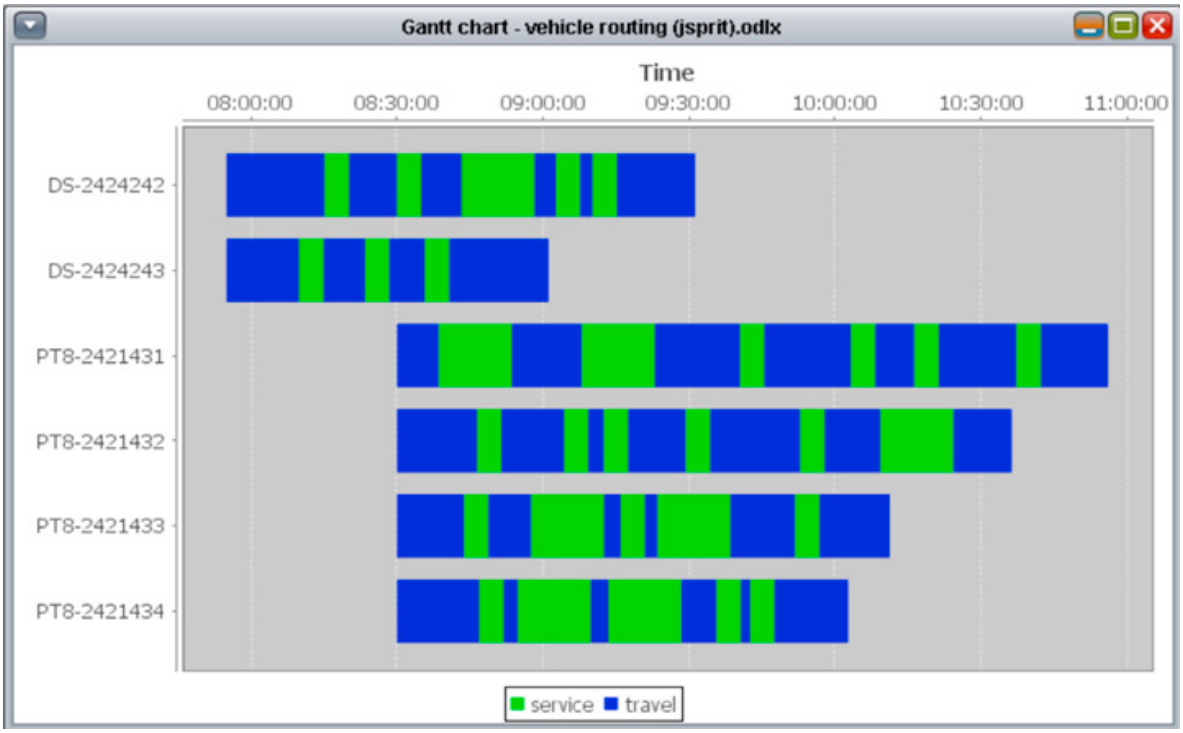

Fig. 5 Daily Gantt Chart of trucks.

Managing fuel consumption relies on monitoring vehicle load variations over time, influenced by route conditions and vehicle weight. Higher engine power and weight increase fuel consumption, emphasizing their role in route planning and vehicle allocation. The ODL VRP tool visually portrays vehicle load changes over time. This graph reveals load variations during collection, offering insights into factors affecting fuel consumption. Decision-makers can pinpoint areas for process enhancement, like route optimization or weight adjustments, by monitoring vehicle load.

The successful implementation of the VRP tool yields significant advantages, notably in terms of fuel savings and reduced social costs. Optimizing routes and minimizing unnecessary trips enhances fuel efficiency, resulting in lower greenhouse gas emissions and environmental benefits.
Moreover, the tool optimizes workforce allocation, reducing labor hours and translating into municipal cost savings. However, many areas face inefficient waste allocation due to inadequate awareness of waste generation. The VRP tool helps identify areas for improvement, enabling cost-effective adjustments and efficient waste collection practices with minimal algorithmic effort and stakeholder involvement.
In summary, the VRP tool's application in waste collection and transportation offers multifaceted benefits, including fuel consumption reduction, cost savings, and improved resource allocation for more efficient waste collection processes.

5. Conclusion

The integration of smart bins and VRP tools offers transformative potential for waste management, promoting sustainability and efficiency while curbing fuel consumption, greenhouse gas emissions, and associated social costs. Optimization results show significant improvements in waste collection and transportation processes, meeting

stakeholder requirements for volume and time constraints. The decision-making capabilities of the VRP tool contribute to effective waste management, underscoring the iterative nature of this approach and the involvement of all stakeholders. Optimizing bin allocation promises substantial cost efficiencies. While the smart bin system and local search heuristic method in ODL have delivered impressive results, future efforts should include deploying more sensors extensively and validating results against a fully operational smart waste collection system. The proven reliability of affordable tools facilitates seamless research deployment. Test various route optimization approaches for deeper insights. Apply the proposed IoT system to different use cases with diverse sensor types, enhancing the credibility of flexible, low-cost IoT tools.

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